

PHYSICAL SCIENCES

High-performance reversible adhesive from PET waste for underwater, structural, and pressure-sensitive applications

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Developing versatile, tough, and sustainable adhesives that function effectively in both wet and dry environments is a major challenge. Here, we report a bioinspired design for versatile, tough, and reversible adhesives upcycled from consumer poly(ethylene terephthalate) (PET) waste. Our approach uses solvent-free, room-temperature dynamic cross-linking of deconstructed PET macromonomers with a diacetoacetate cross-linker, generating a dynamic, vinylogous urethane-bonded, amphiphilic adhesive. Tunable cross-linker concentration and amphiphilicity yield versatile adhesives suitable for underwater, structural, and pressure-sensitive applications on diverse substrates. Our adhesive exhibits high lap-shear strength and work of debonding under both wet and dry conditions, outperforming common commercial adhesives. The dynamic bonds enable thermal repair, on-demand multicycle debonding/rebonding, facile removal, and chemical recycling. Our strategy of transforming plastic waste into versatile, tough, and reversible adhesives offers a sustainable solution for both plastic waste management and next-generation adhesive design while also providing a commercially viable pathway for valorizing plastic waste.

INTRODUCTION

Adhesives play a vital role in our modern world, bonding a variety of materials from everyday items to critical infrastructures (1, 2). Creating versatile, tough, and sustainable high-performance adhesives that can be used across diverse conditions, including wet and dry environments, remains a major challenge. Current adhesive technologies often struggle to meet these multifaceted demands, particularly in terms of adaptability and sustainability (1, 3, 4). One major limitation of most commercial adhesives is instability in underwater conditions, which forces industries to conduct intensive underwater repairs of boat hulls, pipelines, and other undersea infrastructure (e.g., telecommunication and electrical cables) (5, 6). Nature offers inspiring examples of versatile adhesion in wet and dry environments. For example, mussels show strong adhesion to both wet and dry surfaces due to the amphiphilic nature of catecholamine and lysine within their secreted proteins. These amino acids facilitate both covalent and noncovalent interactions with diverse substrates to achieve reversible adhesion (2, 5, 7–17). However, translating these biological mechanisms into synthetic materials has been challenging. Despite extensive research on biomimetic adhesives, which are frequently tailored for wet surfaces, their underwater adhesion strength often falls below 1 MPa (5, 8–11, 13, 15, 16, 18–29). Furthermore, many commonly used structural adhesives, such as epoxies and polyurethanes, are not suitable for underwater adhesion (5, 30–36). Consequently, there is a growing need for versatile, tough, high-performance adhesives, capable of functioning across a wide range of applications in both wet and dry environments.

Traditional adhesives not only lack versatility but also pose environmental concerns due to their irreversible bonding, which makes

them difficult to remove from surfaces, resulting in substantial waste and exacerbated disposal issues at the end of their service life (5, 30–36). Recent advancements have explored the use of dynamic covalent bonds in adhesives, which enhanced sustainability through reversibility and reprocessability under external stimuli (e.g., heat or light) (30, 32, 37–42). However, a critical drawback of most adhesives, including those with dynamic covalent bonds, is their heavy dependence on fossil fuels and other nonrenewable resources, which contribute to carbon emissions and the overall energy footprint. To address this issue, we need well-designed alternative feedstocks for sustainable adhesive development. The use of consumer plastic waste as a source for adhesive building blocks simultaneously reduces dependence on fossil fuels and offers a solution to the global challenge of plastic waste management. Incorporation of plastic waste into upcycled high-performance materials can mitigate the accumulation of polymer waste in oceans, rivers, and seas, where microplastics disrupt ecosystems (37, 43–52). Among the various types of plastic, poly(ethylene terephthalate) (PET) stands out as a major contributor to plastic waste. With an annual production exceeding 70 million tons and an estimated 25 billion tons currently in existence, PET holds substantial potential as a valuable polymer feedstock (53–55). Therefore, we developed a strategy to convert consumer PET waste into value-added materials, such as high-performance reversible adhesives for underwater, structural, and pressure-sensitive applications (Fig. 1), presenting a promising pathway toward a circular economy. To the best of our knowledge, no current technologies have effectively converted PET waste into versatile, tough, and reversible adhesives.

Here, we demonstrated the development of a versatile, tough, reversible adhesive synthesized from consumer PET waste, combining bioinspired amphiphilicity and dynamic interactions (Fig. 1). We designed an amphiphilic tetraamine, synthesized via catalyst- and solvent-free aminolysis of PET using JEFFAMINE T-403 (JAT403), to react with a facially amphiphilic (one side hydrophobic and the other hydrophilic) diacetoacetate cross-linker, generating a dynamic vinylogous urethane-bonded adhesive. The resulting adhesive

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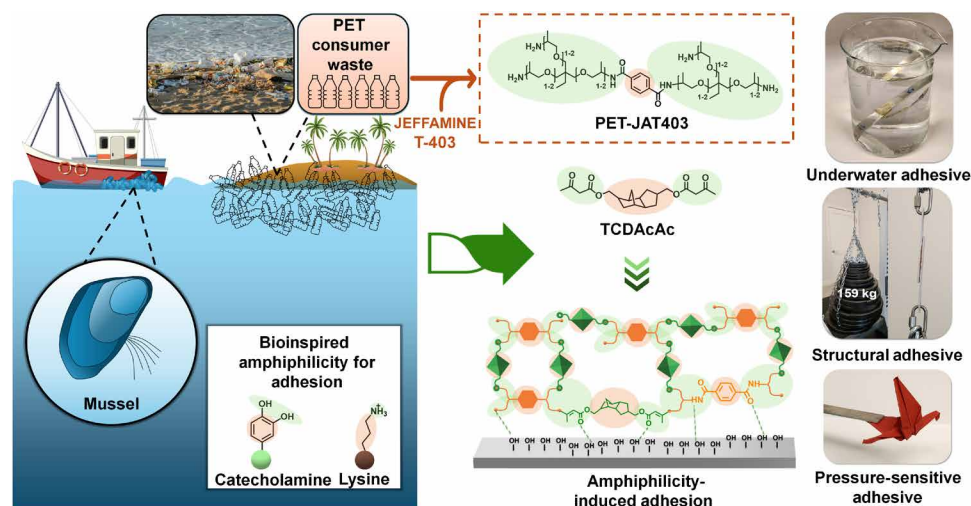


Fig. 1. Design principals of versatile, tough, reversible, and sustainable adhesive. Mussel adhesive proteins, composed of amphiphilic catecholamine and lysine, form robust dynamic interactions with diverse surfaces. The amphiphilic nature of these amino acids is crucial for their adaptability, enabling adhesion in both wet and dry environments. Inspired by this natural system, we have developed a tough, reversible, highly adaptable, and sustainable adhesive from PET consumer waste suitable for underwater, structural, and pressure-sensitive applications. Reproduced image is copyright free and provided by Unsplash.

exhibited strong adhesion to various substrates under diverse conditions, including underwater and dry environments, due to the incorporation of both hydrophobic and hydrophilic moieties. Facial amphiphilicity enhanced adhesion by allowing hydrophilic moieties to promote interfacial bonding, while hydrophobic components strengthened cohesive forces and prevented water infiltration at the adhesive joint, particularly in underwater applications. Moreover, the inclusion of dynamic covalent bonds enabled reversibility, reparability, and recyclability. Analysis of cross-linker stoichiometry revealed that higher cross-linking densities enhanced adhesion under both underwater and dry conditions on common structural substrates, while lower cross-linking densities exhibited pressure-sensitive adhesion. The developed adhesive was room-temperature curable, solvent-free, and applicable in both liquid and dry states, offering high strength and toughness. While this study focused on PET waste and vinylogous urethane bonds, the concept is extendable to other condensation polymers and dynamic amine-based bonds. This approach provides a sustainable route for upcycling plastic waste into a high-performance circular adhesive.

RESULTS

Design of versatile, tough, and reversible adhesive

To design a sustainable, reversible, and versatile adhesive from PET consumer waste, we began by deconstructing PET to form an amphiphilic tetraamine macromonomer. The deconstruction process involved a one-pot, catalyst- and solvent-free aminolysis reaction of solid PET waste with JAT403, a commercially available triamine containing both hydrophilic propylene glycol groups and primary amines (Fig. 2A). The resulting tetraamine macromonomer (PET-JAT403) was isolated via distillation and precipitation in qualitative yield and high purity (figs. S1 and S2). The PET-JAT403 chemical structure features a rigid hydrophobic core and hydrophilic arms, resembling naturally occurring amphiphilic catecholamine adhesives. To maximize amphiphilicity, we designed a facially amphiphilic cross-linker, consisting of a hydrophobic tricyclodecane (TCD) core functionalized

with hydrophilic acetoacetate (AcAc), referred to here as TCDAcAc (scheme S1 and figs. S3 and S4). As a control, a second di-AcAc cross-linker derived from ethylene glycol (EG) was synthesized to understand influences of facial amphiphilicity (scheme S2 and fig. S5). The adhesive, termed TCD-V, was formed by vigorously mixing the amphiphilic PET macromonomer and TCDAcAc cross-linker at room temperature (RT) for 30 s, creating dynamic covalent vinylogous urethane bonds (Fig. 2B and fig. S6A). The liquid state of both the macromonomer and cross-linker at RT provided advantages of surface wettability during application, promoting adhesive strength and eliminating the need for solvents (fig. S6A) (56, 57). The ratio of amine to AcAc functionality will be referenced as the amine:AcAc ratio and is important for ensuring an excess of amines to enable reversibility or reprocessability of the adhesive. Vinylogous urethane exchange occurs via transamination reactions, where free amines within the cross-linked network react with vinylogous urethane bonds when heated above the glass transition temperature (T_g) to reconfigure the polymer network (Fig. 2B). The formation of vinylogous urethane cross-links was confirmed using Fourier transform infrared spectroscopy (fig. S6B), through observation of a reduction in the AcAc C=O peak intensity at 1732 cm^{-1} , indicative of cross-linker consumption. TCD-V showed high storage modulus (G') below T_g and decreased as the T_g was surpassed. The appearance of a slight increase in G' near the end of the rubbery plateau in the dynamic mechanical analysis (DMA) curve (Fig. 2C) further supported the presence of associative-type dynamic cross-linking in TCD-V. This behavior is typical of associative-type dynamic bonds where formation and breakage of the bonds occur simultaneously. This dynamic rearrangement allowed for stress-relaxation behaviors, shown in Fig. 2D, such as those seen in thermoplastics, despite being covalently cross-linked. The characteristic relaxation times (τ) were determined using both stretched exponential fits (Kohlrausch fits) and a simplified "1/e" approach. The corresponding τ obtained from Kohlrausch fits interpolated using Arrhenius law yielded the activation energy of the vinylogous urethane exchange in the 10:5 TCD-V with a value of 275 kJ/mol (Fig. 2E). This Arrhenius behavior confirmed

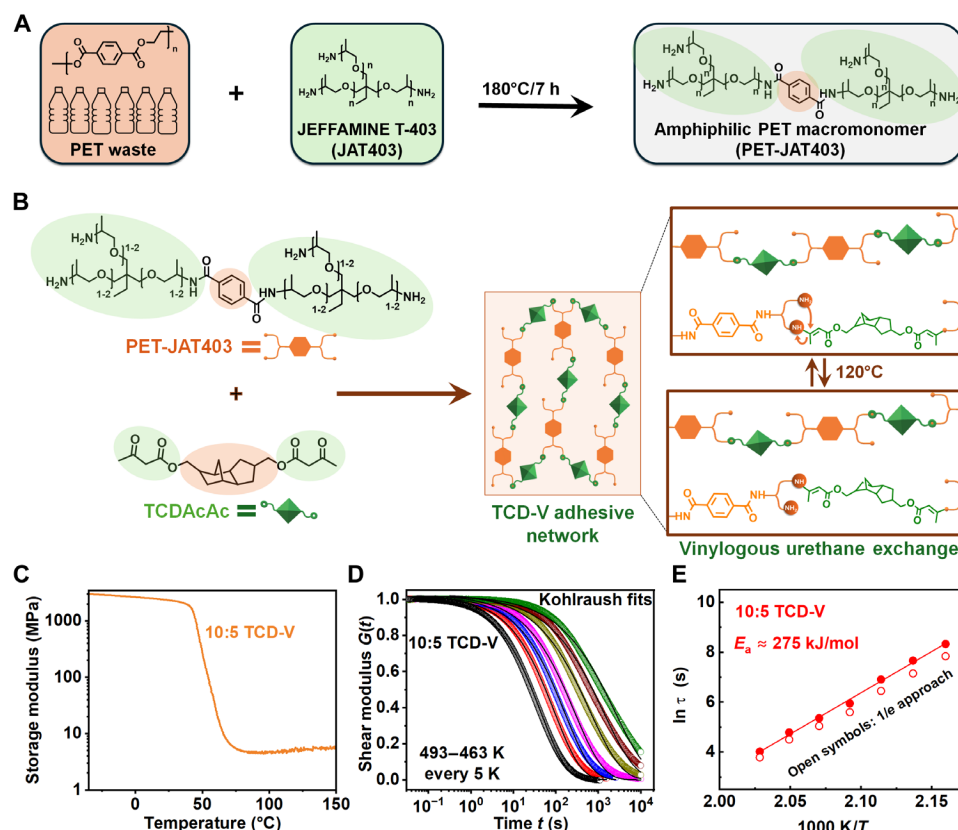


Fig. 2. Synthesis of amphiphilic PET macromonomer and reversible adhesive from PET waste. (A) Scheme showing the deconstruction of consumer PET via aminolysis with JAT403 to yield an amphiphilic, tetraamine macromonomer. h, hours. (B) Combining the PET-JAT403 and TCDAcAc yielded a cross-linked network held together through vinylogous urethane bonds. Vinylogous urethane networks undergo thermally induced transamination exchange reactions (associative exchange), where a free amine within the polymer network reacts with the vinylogous urethane bond, allowing for reversibility and reprocessability. (C) DMA of bulk 10:5 TCD-V revealed a high storage modulus below the T_g , a sharp decrease above T_g , and a well-defined rubbery plateau at higher temperatures, indicative of a cross-linked network. (D) Stress relaxation behaviors of 10:5 TCD-V over a temperature range of 493 to 463 K with an interval of 5 K. The shear modulus decreased at an accelerating rate with increasing temperature. (E) Analysis of the relaxation time for the 10:5 TCD-V sample (red) as a function of inverse temperature revealed a linear Arrhenius relationship, yielding an activation energy (E_a) of 275 kJ/mol.

the reversibility of the vinylogous urethane exchange, indicating that the adhesive exhibited vitrimer-like characteristics and their relaxation process was thermally activated. The cross-linking was further validated by its resistance to various solvents, including water, tetrahydrofuran, acetone, and toluene, after 16 hours of submersion (fig. S6C). In this solvent resistance study, no notable dissolution was observed, even after 72 hours of submersion in water. Subsequent characterization of the swelling ratio and soluble fraction of the cross-linked network in water for 72 hours revealed a 0% swelling ratio and a minimal soluble fraction (2.9%) for the 10:7 TCD-V adhesive (fig. S6D), confirming the underwater stability of the TCD-V adhesive. We hypothesized that the balanced amphiphilicity in the TCD-V would enable strong adhesion to diverse substrates under both wet/underwater and dry conditions through noncovalent interactions such as van der Waals and hydrogen (H) bonding.

Underwater adhesive performance

Underwater adhesion of 10:7 TCD-V on aluminum (Al) substrates was evaluated first. The viscous TCD-V was applied directly to submerged substrates using a syringe and joined together by applying hand pressure (Fig. 3A), demonstrating immediate adhesion. The

specimens were cured underwater at RT ($\sim 18^\circ\text{C}$) for 16 hours under applied pressure ($\sim 800\text{-g}$ weight) before lap-shear testing. After curing, lap-shear strength was measured, and representative force-extension curves, shown in Fig. 3B, revealed substantial underwater adhesion (6.51 MPa). To optimize the curing time, we joined a series of Al substrates underwater with the 10:7 TCD-V adhesive and allowed them to cure at RT for 10 min and 1, 2, 4, 6, and 16 hours before lap-shear testing. The results (Fig. 3C) revealed that a minimum of 2 hours was required to achieve high adhesion strength. Beyond this point, all samples exhibited comparable lap-shear strength. Next, we investigated the effect of dynamic cross-linking density on the underwater adhesion performance of TCD-V adhesive on Al substrates. The tetraamine macromonomer allowed stoichiometric control of cross-linking density by varying the amount of TCDAcAc cross-linker. We hypothesized that increased TCDAcAc concentration and, thus, higher hydrophobicity and cross-linking would enhance both bulk network integrity and underwater adhesion through water repellence. To test this, we bonded Al substrates underwater with varying amine:AcAc ratios (10:2, 10:3, 10:5, and 10:7). Underwater adhesion increased with the amine:AcAc ratio, where the 10:7 TCD-V exhibited the highest lap-shear strength (6.51 MPa) on Al

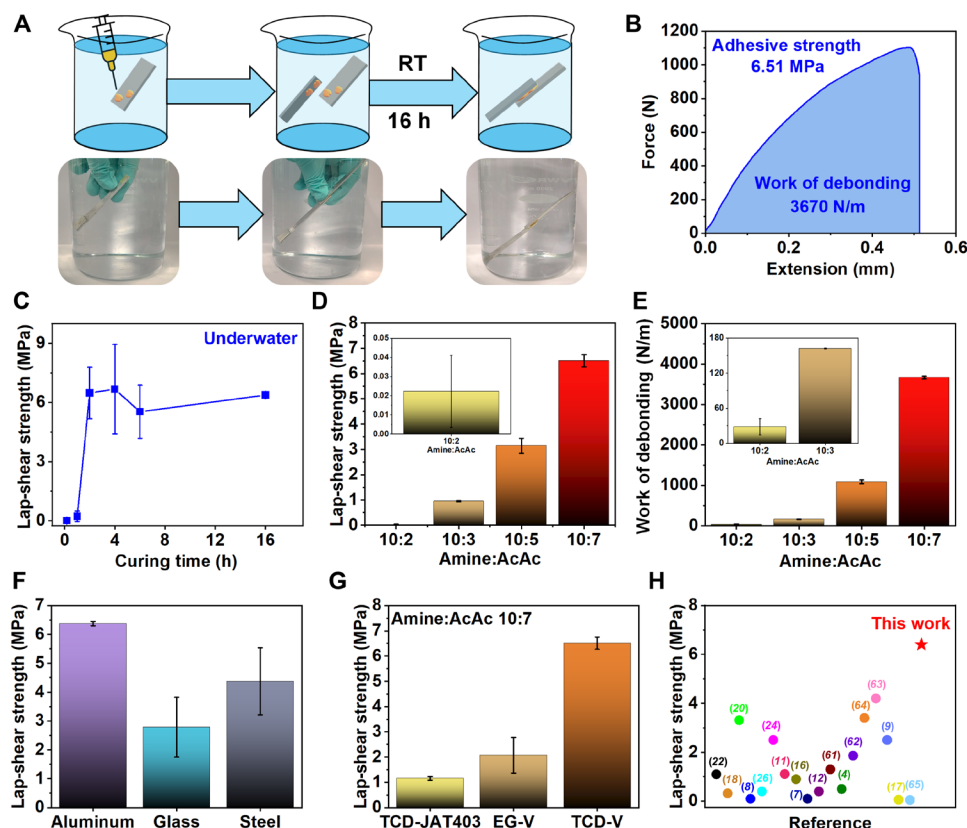


Fig. 3. Underwater adhesive performance of TCD-V. (A) Adhesive was applied to the substrates, and they were joined together underwater. The assembled joint, self-supporting immediately after bonding, was allowed to cure underwater for 16 hours at RT. (B) Force-extension curves showing the underwater adhesive performance of the 10:7 TCD-V after curing for 16 hours in RT water. (C) Lap-shear strength of the 10:7 TCD-V relative to curing time. (D) Lap-shear strengths for varying amine:AcAc of the TCD-V after underwater curing at RT. The 10:7 formulation showed the highest adhesion with a lap-shear strength of 6.51 MPa. (E) Work of debonding values for various amine:AcAc ratios of the TCD-V adhesive. (F) Lap-shear strengths for underwater adhesion of TCD-V to different substrates. Adhesive strength was directly correlated with the hydrophilicity of the surface where less hydrophilic substrates showed greater lap-shear strengths. (G) Force-extension curves for the 10:7 TCD-V and controls where PET-JAT403 was replaced with JAT403 (TCD-JAT403) and TCDAcAc was replaced with EGAcAc (EG-V). (H) Plot comparing recent examples of underwater adhesives from the literature compared to our adhesive. For a more direct comparison, references were limited to those conducted on metal with short (≤ 24 hours) curing times. Our work surpasses all the examples documented here (4, 7–9, 11, 12, 16–18, 20, 22, 24, 26, 61–65). All error bars represent the SD ($n = 3$). h, hours.

(Fig. 3D). Conversely, 10:2 TCD-V showed minimal adhesion (22.2 kPa) (Fig. 3C and table S1). To quantify adhesive toughness, we calculated the work of debonding from the force-extension curves in the lap-shear experiments (Fig. 3E and fig. S6E). The integrated area underneath the force-extension curve is defined as the work of debonding, or work of adhesion, describing the energy required to break the adhesive joint. For the underwater samples, the surface area varied but maintained $\sim 144 \text{ mm}^2$ on average, following the procedure from the American Society of Testing Measurement (ASTM) D1002. Following the same trend as the lap-shear strength, the 10:7 exhibited the highest value for work of debonding (3670 N/m) and 10:2 the lowest (Fig. 3E and table S1). The highly cross-linked and fully cured TCD-V adhesive network likely inhibited water permeation, enhancing cohesive strength and, thus, underwater adhesion. Failure mode analysis supported this observation; fig. S7 showed adhesive failure for 10:7 TCD-V and cohesive failure for 10:2 TCD-V, indicating a weaker polymer network in the latter. A similar experiment was conducted to determine underwater adhesion in marine environments where the 10:7 TCD-V was cured at RT for 16 hours in a solution of simulated seawater (3.5% NaCl). Although adhesion

was diminished relative to the samples cured in deionized water, it still exhibited strong underwater adhesion with a lap-shear strength of 3.54 MPa (table S1).

Having established the influence of dynamic cross-link density and hydrophobicity of the TCD-V adhesive on lap-shear strength, we also sought to identify the impact of substrate hydrophilicity on underwater adhesion. Two additional substrates with different wettabilities were chosen, in this case, untreated steel and glass, and compared to the results from Al substrates. Glass had the highest wettability with a water contact angle of $\sim 27^\circ$ (58), followed by steel at $\sim 51^\circ$ (59, 60) and Al at $\sim 70^\circ$ (59, 60). Using the highest-performing 10:7 TCD-V adhesive formulation, we observed that glass, as anticipated, exhibited the lowest, although still high, adhesion (2.79 MPa) (table S2). This result was likely due to rapid interaction between the glass and the surrounding water, forming a hydration layer on the substrate surface, which hindered adhesion. Following the trend of substrate hydrophilicity, adhesion to steel (4.37 MPa) was greater than glass but weaker than the more hydrophobic Al (Fig. 3F). To validate the facial amphiphilic design of our adhesive, we synthesized two control formulations: TCD-JAT403 and an ethylene glycol

vitriimer (EG-V), both using a 10:7 amine:AcAc ratio (Fig. 3G). In TCD-JAT403, the amphiphilic PET macromonomer was replaced with JAT403 to assess the role of the terephthalamide core. In EG-V, TCDAcAc was substituted with EGAcAc to evaluate the importance of the facially amphiphilic TCDAcAc component. The lap-shear adhesion results revealed that the TCD-JAT403 exhibited weaker underwater adhesion strength (1.01 MPa; Fig. 3G). EG-V demonstrated improved underwater adhesion (2.07 MPa) compared to TCD-JAT403, yet it was still weaker than the 10:7 TCD-V adhesive (6.51 MPa). These results confirmed that both amphiphilic PET-JAT403 and the facially amphiphilic TCDAcAc are essential for achieving strong underwater adhesion. A comparison of the underwater lap-shear strength on Al with literature values demonstrated that 10:7 TCD-V outperformed reported adhesives (Fig. 3H) (4, 7–9, 11, 12, 16–18, 20, 22, 24, 26, 61–65).

Strong and tough adhesion for structural applications

Beyond the underwater adhesive properties of TCD-V, we also investigated its performance under dry conditions, particularly for structural applications. Initially, Al substrates and 10:7 TCD-V were used to investigate the effects of curing time and temperature. The TCD-V adhesive was applied to Al substrates by spreading the mixture evenly across the surface, such that ~10 mg of adhesive was applied over a 144-mm² area. After handpressing, the adhesive joint was cured at RT for 16 hours before testing the lap-shear strength (Fig. 4A). The resulting lap-shear strength (8.48 MPa) of TCD-V demonstrated its potential as an RT-curable adhesive. To further optimize the formulation for RT curing, we repeated the same experiment under controlled pressure at 10 psi (68.9 kPa) using a binder clip. Under these conditions, strong adhesion was achieved after only 4 hours of RT curing, yielding a lap-shear strength of 17.1 MPa (fig. S8). This enhancement was likely due to the applied pressure, promoting better surface wetting and facilitating more noncovalent interactions at the adhesive-substrate interface. Building on the high strength of the RT-curable adhesive, we explored the impact of higher-temperature curing. Curing above the T_g promoted optimal rearrangement of the dynamic vinyllogous urethane bonds, leading to stronger noncovalent interactions with hydroxy-terminated surfaces. To determine the optimal curing conditions, we first measured the T_g of the 10:7 TCD-V adhesive, which was found at 63.5°C (fig. S9). Subsequently, Al substrates bonded with the adhesive were cured at 80°C for varying durations (1, 2, 4, 6, and 16 hours) and then tested for lap-shear strength at RT (Fig. 4B). Samples cured for 1 hour achieved a lap-shear strength of 10.5 MPa, while those cured for 6 hours yielded 10.9 MPa. Both values are within the margin of error of the 16-hour result (12.1 MPa), indicating efficient and fast curing. While overall lap-shear strength remained similar, failure mode appeared to be influenced by curing time: Shorter times resulted in more adhesive failures, while longer times showed more cohesive failures (fig. S10). Presumably, at short curing times, the reaction between macromonomer amines and AcAcS proceeded faster than establishment of substrate interactions, leading to adhesive failure. Conversely, longer curing times allowed greater rearrangement of interactive groups and more complete curing within the adhesive, enhancing substrate adhesion and resulting in cohesive failure.

The influence of dynamic cross-link density on adhesion to Al substrate was investigated by varying the amount of TCDAcAc cross-linker. TCD-V adhesives with a series of amine:AcAc ratios, 10:1, 10:2, 10:3, and 10:5, were tested on Al after curing at 80°C for 6 hours.

The lap-shear adhesion strength trends observed under dry conditions were opposite to those observed for underwater adhesion. For example, the 10:3 and 10:5 formulations exhibited the highest lap-shear strengths (13.6 and 13.4 MPa, respectively; Fig. 4C and table S3), whereas the 10:7 formulation yielded 12.1 MPa. This suggests that the higher concentration of unreacted primary amines in the 10:3 and 10:5 TCD-V formulations promoted stronger hydrogen-bonding interactions with the substrate, as well as within the vitriimer network, leading to enhanced adhesion. Furthermore, 10:3 and 10:5 TCD-V showed lower T_g values (45.3° and 58.4°C, respectively), which likely facilitated efficient network rearrangement at the interface, further enhancing noncovalent interactions and wettability (fig. S9). Conversely, the lower concentration of free amines and higher T_g of the 10:7 TCD-V (63.5°C) adhesive may have hindered noncovalent interactions and network rearrangement under dry conditions.

Analogous to our underwater adhesion study, we also investigated the role of facial amphiphilicity on structural adhesion performance. To assess the contribution of the terephthalamide core, we tested a control adhesive (TCD-JAT403; 10:7 amine:AcAc) on Al substrates which showed reduced lap-shear strength (4.61 MPa) compared to TCD-V, despite identical amine functionality (Fig. 4C). This underscored the importance of the hydrophobic terephthalamide moiety in enhancing rigidity and cohesive strength. Another control adhesive, EG-V, designed to specifically evaluate facial amphiphilicity, demonstrated improved adhesion (10.8 MPa) compared to TCD-JAT403 yet still did not match the superior performance of 10:7 TCD-V (Fig. 4C). Collectively, these findings emphasized that both PET-JAT403 and TCDAcAc components were essential for optimal adhesive strength and toughness. The force-extension curves (Fig. 4D) showed a sharp initial rise, followed by a gradual increase in force until failure, indicating ductile, plastic behavior. This strain-hardening characteristic is rare among adhesives. The work of debonding was calculated from force-extension curves (Fig. 4D) and was greatest for the 10:2 TCD-V (9010 N/m), nearly six times higher than commercial J-B Weld epoxy (1569 N/m) (fig. S11). Most existing commercial structural adhesives are brittle, with low work of debonding values. The soft polypropylene oxide units of the PET macromonomer allowed for higher extensibility and an overall tougher adhesive, as they could dissipate mechanical stress throughout the joints and prevent abrupt failure. This enhanced toughness contributed to the high load-bearing capacity of these adhesives (Fig. 1), enabling them to mitigate premature adhesive failures and making them highly desirable for structural applications. Notably, the lap-shear strength of our adhesive outperforms current literature benchmarks (fig. S12) (14, 42, 63, 66–76).

The TCD-V adhesive demonstrated versatile adhesion to various hydroxylated surfaces, including common structural substrates such as steel, wood, and glass. The same methodology used for the Al tests was applied to these substrates, using the 10:5 TCD-V, given its exceptionally high work of debonding while maintaining a high lap-shear strength. Lap-shear tests from steel surpassed that of the Al substrates with an ultimate strength of 14.9 MPa and a work of debonding equal to 7780 N/m (Fig. 4E and table S4). For glass, the procedure was modified to avoid premature breakage of the glass substrate by reducing the working area from 144 to 36.0 mm². Even so, substrate failure was still an issue—the glass shattered well in advance of damage to the adhesive joint but not before exhibiting lap-shear strengths upward of 20.0 MPa (fig. S13 and table S4). This performance was likely due to the high density of hydroxyl groups

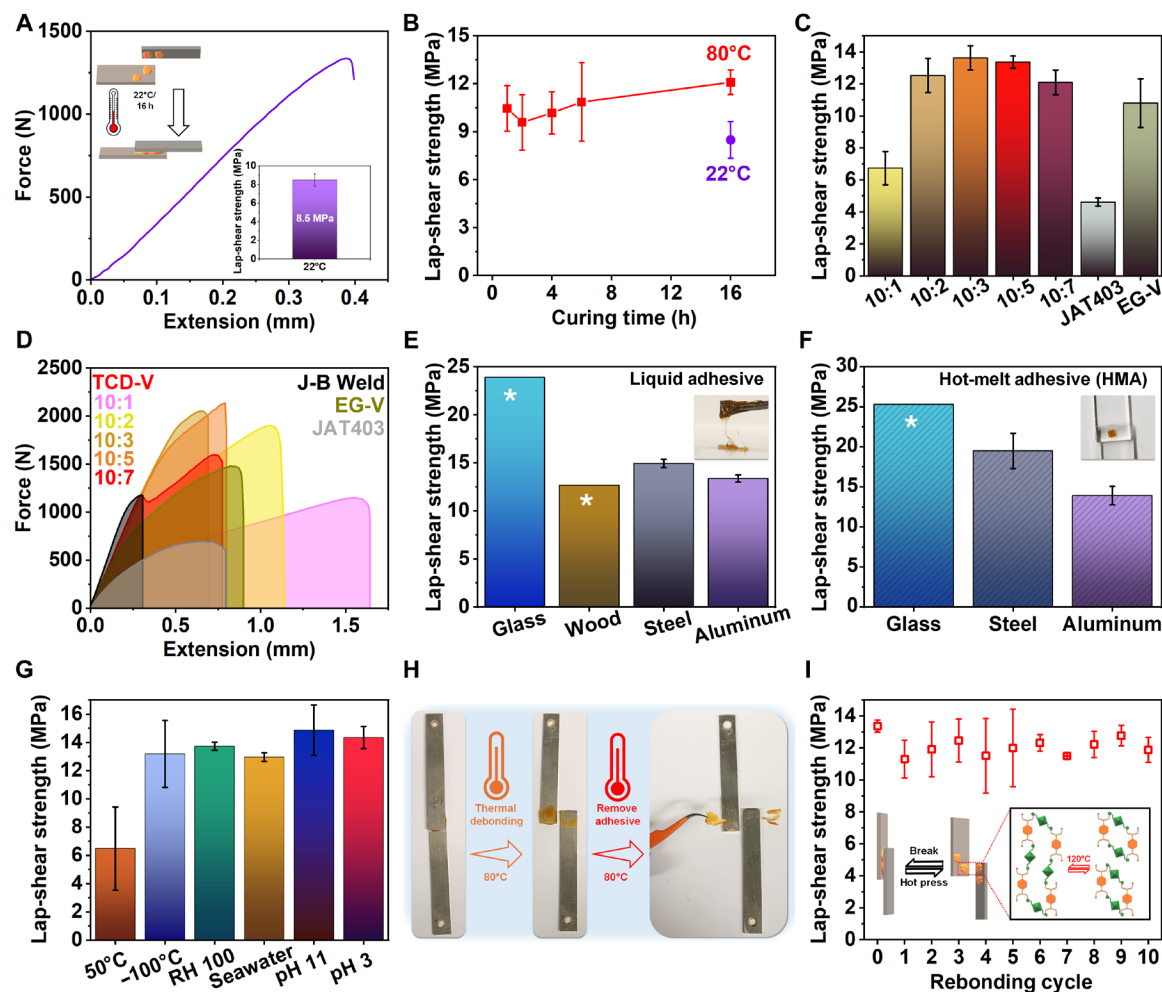


Fig. 4. Tough and reversible TCD-V adhesive's performance on different structural substrates. (A) The force-extension curve for the 10:7 TCD-V adhesive cured at RT shown with the average lap-shear strength. (B) Plot showing lap-shear strength as a function of curing time at 80°C (red). We also plot the RT curing lap-shear strength data (purple) to visualize the difference between 80°C and RT. (C) Bar graph showing lap-shear strengths for varying amine:AcAc ratios of the TCD-V adhesives compared with a 10:7 JAT403 control (JAT403) and a 10:7 EGAcAc control (EG-V). (D) Force versus extension curves demonstrated strong-ductile (tough) adhesion. The shaded area is marked as the area used to calculate work of debonding. (E) Lap-shear strength for the liquid 10:5 TCD-V adhesive for four different substrates. Those marked with a white asterisk showed substrate failure and are therefore not indicative of the adhesive strength. (F) Lap-shear strength for the dry film of 10:5 TCD-V adhesive [hot-melt adhesive (HMA)] for three different substrates. (G) Lap-shear strength for the 10:5 adhesive in extreme environments is shown. Relative humidity of 100% (RH 100) describes the results from the substrate being fully submerged in water for 24 hours. (H) The image showing thermal debonding and facile removal of adhesive from the surface at 80°C. (I) Rebonding ability tests for 10:5 TCD-V adhesive. The plot shows no significant change in lap-shear strength even after 10 adhesion cycles. The cartoon inset shows the process for re-adhesion (hot pressing at 120°C). All error bars represent the SD ($n = 3$). h, hours.

inherent to the chemical structure of borosilicate glass. This experiment was repeated with the original testing area (144 mm^2) on sanded wood substrates. The lap-shear testing again resulted in substrate failure, where the wood, like the glass, sheared off near the adhesive joint following the grain line (fig. S13). In this case, the strong adhesion was likely due to the high density of hydroxyl groups in cellulose and increased wettability granted by the porosity of the wood. Given the observed strong adhesion to various hydroxyl-containing surfaces, we investigated whether this adhesion could be maintained between dissimilar substrates—a crucial requirement in automotive and aerospace applications, where joining composites to metals such as aluminum or steel presents notable challenges. We joined both glass fiber- and carbon fiber-reinforced polymers (GFRPs and CFRPs, respectively) to Al using the 10:5 TCD-V adhesive and

measured lap-shear strengths. Both joints exhibited remarkably high lap-shear strengths (11.1 and 15.7 MPa for the GFRP and CFRP, respectively) (fig. S14). The solid film of TCD-V was also used as a hot-melt adhesive (HMA) on various substrates (e.g., Al, steel, and glass). At temperatures above the T_g , the TCD-V film underwent continuous bond exchange/cleavage, resulting in reduced viscosity and promoting wetting of the substrate. The lap-shear strength of the solid adhesive was measured using 6 mm by 6 mm of the 10:5 TCD-V for aluminum and steel substrates and 2 mm by 2 mm for the glass substrates. Wood was not used for this experiment given its poor thermal conductivity. The pieces of TCD-V were placed between the substrates and pressed at 120°C and 500 psi (3447 kPa) for 1 hour on a Carver press. The HMA performed in alignment with the trend established by the liquid adhesive, with steel (19.5 MPa)

exceeding aluminum (13.9 MPa) (Fig. 4F and table S5). Similar to the liquid adhesive, the glass exhibited structural failure well before adhesive failure despite displaying a high lap-shear strength (>25 MPa) (Fig. 4F and table S5). The HMA was robust for all tested substrates and implied that both methods of application (liquid or solid) are viable for the TCD-V adhesive.

To assess adaptability to harsh environmental conditions, we evaluated the performance of the TCD-V adhesive across a wide temperature range. We measured the lap-shear strength of 10:5 TCD-V adhesive at 50°C on Al substrates and achieved at 6.48 MPa (Fig. 4G and fig. S15), slightly lower than that at RT value, likely due to the proximity of the testing temperature to the measured T_g (63.5°C) of 10:5 TCD-V (fig. S9). At much lower (-100°C) temperatures, the adhesive maintained a high lap-shear strength of 13.2 MPa even after a 2-min exposure to liquid nitrogen. This resilience at low temperatures is likely due to the high modulus of TCD-V at -50°C (Fig. 1C). Furthermore, the adhesive demonstrated excellent durability under extreme humidity. For example, after 2 days of submersion in water at RT [relative humidity at 100% (RH 100)], the lap-shear strength of Al substrates remained unchanged at 13.7 MPa (Fig. 4G and table S6). With the promising results from the submerged samples, we explored more extreme conditions including corrosive solutions of simulated seawater (13.0 MPa), base (pH 11; 14.9 MPa), and acid (pH 3; 14.3 MPa). Notably, strong adhesion was maintained across all tested extreme environmental conditions (Fig. 4G and table S6). These results highlight the TCD-V adhesive's resilience and versatility, making it a promising candidate for applications demanding high performance and stability in harsh environments.

The dynamic covalent vinylogous urethane and noncovalent bonds (hydrogen bonding and van der Waals interactions) within TCD-V enable reversible macroscopic responses to thermal stimuli, making it a promising reusable structural adhesive. Unlike most commercial adhesives, TCD-V can heal broken joints upon heating above its T_g , which allowed for easy removal of adhesive from the substrates at 80°C (Fig. 4H). While most adhesives require extreme temperatures for removal either through thermal degradation or heating above their T_g , our adhesive enables facile detachment and rebonding under moderate conditions, facilitated by dynamic vinylogous urethane bond exchange, thus providing a practical and energy-efficient alternative. The rebonding capability of the TCD-V was assessed through a cyclic sequence of breaking the lap joint of Al substrates at RT and subsequent healing of the joint at elevated temperature (120°C) and applied pressure [500 psi (3447 kPa)] (Fig. 4I). The average lap-shear strength remained above 11.5 MPa over 10 cycles, demonstrating consistent performance despite minor variations likely attributed to inconsistencies in pressing small material amounts and the presence of imperfections (e.g., dust, bubbles, and detritus). This sustained strength demonstrates a substantial advantage over traditional adhesives. Furthermore, the TCD-V adhesive was completely chemically recyclable, where the PET-JAT403 macromonomer recovered by cleaving the vinylogous urethane bond with excess amine (fig. S16). To further evaluate the efficacy of our high-performance reversible adhesive for practical applications, we compared its lap-shear strength to that of commonly used commercial adhesives, including J-B Weld (epoxy), Loctite 416 and Loctite Professional Liquid (both ethyl cyanoacrylates), and Gorilla Glue (polyurethane) (fig. S17). Both the 10:5 TCD-V liquid and HMA formulations outperformed all four commercial adhesives on Al substrates (144-mm² area; cured according to the manufacturer's instructions) (fig. S17) (30).

Pressure-sensitive adhesive application

Having demonstrated the tunability of the TCD-V adhesive for both underwater and structural applications by controlling the degree of cross-linking, we investigated its potential for pressure-sensitive applications. Pressure-sensitive adhesives (PSAs) are commonly used for temporary bonding, requiring rapid and relatively weak adhesion, as exemplified by adhesive tapes, bandages, and sticky notes. PSA adhesion relies on pressure-induced noncovalent surface interactions without the need for heat or irradiation. The adhesion of PSAs is aided by their low T_g , which facilitates surface wetting at RT. We hypothesized that low cross-linking densities would decrease T_g and increase free amine density, promoting pressure-sensitive adhesion through noncovalent interactions with hydroxyl surfaces. To test this hypothesis, we reacted the PET macromonomer with the TCDAcAc in a 10:0.5 amine:AcAc ratio by mixing the two components together in a scintillation vial and curing for 1 hour at 80°C to complete cross-linking. The resulting viscous gel showed a low T_g of 11.8°C, making it a potential candidate for PSA under ambient conditions (fig. S8). Subsequently, a series of experiments were completed where the adhesive was applied over a surface area of 144 mm² on an Al substrate and was attached by hand to several different materials. The 10:0.5 TCD-V showed adhesion to all tested substrates (Teflon, paper, polycarbonate, glass, steel, and wood) and was later easily removed from the surfaces (Fig. 5A). To quantify this adhesion, substrates were prepared out of each material and overlapped to form an adhesive joint of ~ 144 mm² with the 10:0.5 TCD-V adhesive before pressing for 10 min with an 800-g weight. After the allotted time, the specimens were tested for load-bearing capacity and lap-shear strength (Fig. 5, B and C). All substrates showed reversible adhesion on the kilopascal scale. The highest adhesion was observed for the Al, steel, and glass substrates with lap-shear strengths of 50.2, 45.3, and 43.7 kPa, respectively (table S7). PSAs ideally exhibit reusability, reforming bonds after joint failure. The rebonding capability of 10:0.5 TCD-V on aluminum substrates was evaluated through repeated lap-shear testing. After breakage of the joint, the substrates were realigned and weighted with an 800-g weight for 10 min to maximize adhesion before testing for lap-shear strength. Over 10 cycles, no adhesion loss was observed (Fig. 5D), demonstrating that the TCD-V adhesive is a promising candidate for PSA applications. To investigate the influence of high-pressure environments on TCD-V adhesion, we conducted a secondary experiment using a Carver press at 145 psi (1000 kPa) (fig. S18). The resulting adhesion was greatly increased, reaching a maximum of 563 kPa. However, this enhanced adhesion diminished after 4 cycles, potentially due to adhesive leakage from the joint, resulting in reduced adhesive residue on the bonding surfaces.

After the investigation of pressure-sensitive adhesion, we conducted a preliminary life-cycle analysis (LCA) to assess the adhesive's sustainability. Details regarding the methodology for the LCA are included in the Supplementary Materials. The LCA was conducted using model compounds to represent the benchtop synthesis. Results from the analysis revealed carbon footprints of less than 3 kg of carbon dioxide (CO₂)/kg for both PET-JAT403 and TCDAcAc (2.29 and 2.68 kg of CO₂/kg, respectively). These values are favorably low compared to typically reported reagents used in synthetic chemistry (figs. S19 and S20 and tables S8 and S9). Further modeling of the adhesive's carbon footprint resulted in a value of 2.39 kg of CO₂/kg for the TCD-V, representing a great improvement over similar structural adhesives currently in use (figs. S21 and S22 and tables S10 and S11).

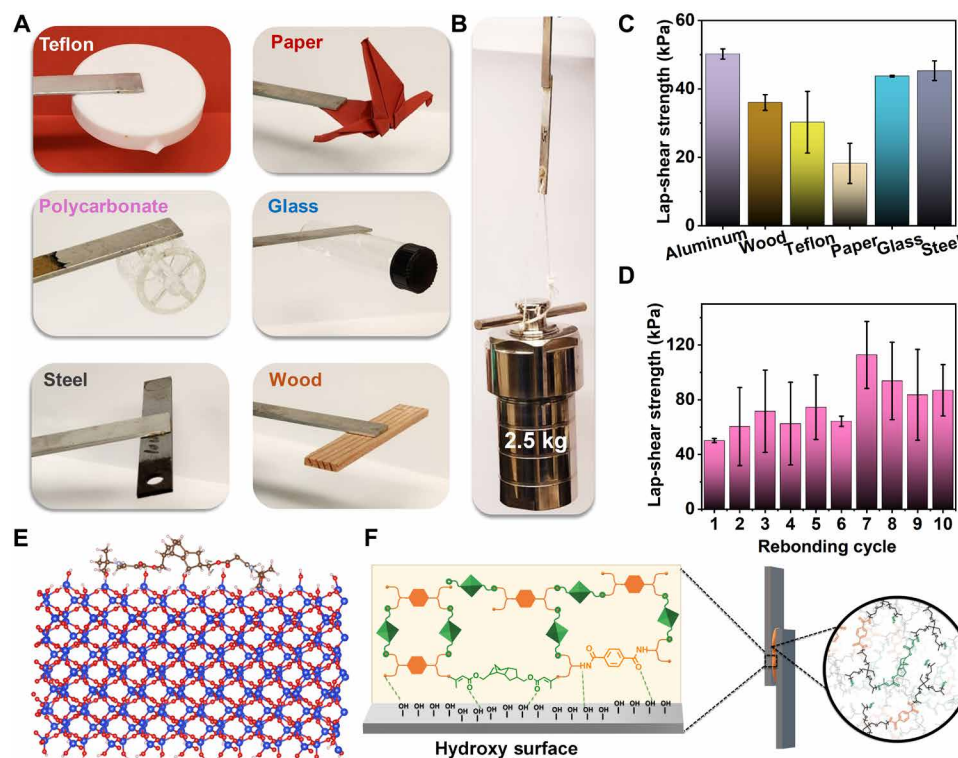


Fig. 5. PSA and overall mechanistic investigation. (A) Images showing pressure-sensitive adhesion of TCD-V (10:0.5) with Teflon, paper, polycarbonate, glass, steel, and wood. (B) Image showing a load-bearing capability where a 144-mm² lap joint of the TCD-V PSA holds a 2.5-kg metal weight. (C) Lap-shear strength of the TCD-V PSA on six different common substrates. The strongest adhesion was seen on aluminum with a lap-shear strength of 50.2 kPa. (D) Lap-shear strength of PSA over 10 cycles of breaking and repressing of the lap joint. Adhesion was maintained with no losses over a period of 10 cycles. (E) Molecular structure of a TCDAcAc model molecule on a model silica surface. DFT calculations confirmed that multiple hydrogen bonds can form between TCD-V and hydroxylated surfaces. (F) Cartoon showing the proposed mechanism for adhesion between the TCD-V network and the substrates. Hydroxyl groups on the surface of the substrates can interact with carbonyl, amide, and primary amines of the TCD-V through hydrogen bonding. All error bars represent the SD ($n = 3$).

Mechanism for tough reversible adhesion

To further elucidate the underlying mechanism that enables the TCD-V adhesive to have such versatile, tough, and reversible adhesion, we investigated its adhesive properties using density functional theory (DFT) calculations (Fig. 5E and figs. S23 to S29) (77–90). For DFT calculations, a model of hydroxylated silica was used to mimic the hydroxy substrates. DFT calculations revealed that TCD-V can form intermolecular and intramolecular hydrogen bonding using several potential hydrogen-bonding interaction sites between TCD-V and the substrate, including primary amines, amides on the terephthalamide core, and esters from the AcAc cross-linkers. These calculations further indicated that the tricyclic ring of the TCDAcAc cross-linker preferentially orients carboxyl groups toward hydroxylated surfaces at a distance of ~ 1.7 Å, (Fig. 5E), suggesting facial amphiphilicity. For each TCDAcAc unit near a surface, two moderately strong hydrogen bonds were formed and contributed to a binding energy of 115.3 kcal/mol (figs. S23 and S24). Additional DFT calculations for interactions between the terephthalamide core and hydroxylated surfaces were performed, accounting for N–H, N–H₂, and C=O bonds present in the amphiphilic macromonomer structure (figs. S25 to S28). These functional groups were also found to form strong H-bonding interactions with the simulated surfaces, exhibiting an overall large binding energy of 187.1 kcal/mol, consistent with the high adhesion of the TCD-V. The strong binding

energies explained the tough adhesive nature, but achieving reversibility required breaking these bonds, which occurred at elevated temperatures ($\sim 80^\circ$ to 120°C). Furthermore, the presence of multiple dynamic hydrogen bonds and vinylogous urethane bonds explained the retention of adhesion even after 10 thermal reprocessing cycles. Upon heating, the TCD-V network reorients, aligning electron donors toward hydroxylated substrate surfaces and facilitating dynamic bond exchange and reformation. This realignment reestablished key physical interactions identified by DFT, including hydrogen bonding, metal-ion coordination with oxygen, and van der Waals forces, thereby maximizing adhesive strength. The proposed mechanism of adhesion involves synergistic contributions from cohesive and adhesive interactions. Specifically, hydrogen bonding and vinylogous urethane dynamic covalent bonding within the TCD-V adhesive significantly enhance its cohesive strength. Concurrently, noncovalent interactions—including hydrogen bonding, metal-oxygen coordination, and van der Waals forces—between hydroxyl-terminated substrates and the TCD-V adhesive strengthened interfacial adhesion (Fig. 5F). Collectively, these dynamic physical and chemical bonds facilitated the surface contact at macroscopic and microscopic scales and improved both cohesive and adhesive forces, leading to enhanced adhesive strength.

To investigate the underwater adhesive properties of TCD-V, we used a continuum water model to compute the binding energy

under aqueous conditions (see the Supplementary Materials for details) (91). DFT calculations showed that the underwater binding energy of TCD-V to the surface decreased (~ 47.3 kcal/mol) compared to vacuum conditions, and the distance between TCD-V and the surface increased slightly ($\text{C}=\text{O}\cdots\text{H}$ distance of ~ 1.73 Å). Despite this reduction, the underwater adhesive performance remained relatively strong (fig. S29). The adhesion mechanism governed by hydrogen bonding remained similar to vacuum conditions; however, these interactions were partially screened by water, weakening their overall effect. These findings aligned with experimental observations where more hydrophobic surfaces (e.g., Al) exhibited stronger adhesion than hydrophilic ones (e.g., glass). The strong adhesion, reprocessability, and versatility of TCD-V highlight the potential of dynamically bonded networks for advanced adhesive design.

DISCUSSION

In this study, we introduced a bioinspired design strategy based on facial amphiphilicity and dynamic interactions to develop a versatile, tough, and reversible adhesive derived from postconsumer PET waste. This approach addressed a long-standing challenge in adhesive technology: achieving strong, reversible adhesion in both underwater and dry environments, while maintaining sustainability and recyclability. Our adhesive system was constructed from a PET-derived amphiphilic macromonomer featuring a hydrophobic terephthalamide core and hydrophilic, amine-terminated propylene glycol arms, combined with a facially amphiphilic diacetoacetate cross-linker to form dynamic vinylogous urethane bonds. This solvent-free and RT-curable formulation yielded a robust adhesive network with local amphiphilicity, enabling strong interfacial interactions. The resulting adhesives demonstrated high performance across diverse applications—from underwater and structural to pressure-sensitive adhesion. Notably, the 10:7 TCD-V achieved an underwater lap-shear strength of 6.51 MPa on Al, outperforming most literature-reported underwater adhesives. This enhanced underwater performance was attributed to the hydrophobic, rigid terephthalamide core and the facially amphiphilic tricyclic structure, which minimized water ingress and preserved network integrity under wet conditions. Under dry conditions, our adhesives maintained strong adhesion to a wide range of structural substrates, including glass, Al, steel, wood, and even dissimilar substrate pairings such as metal-composite interfaces. Under extreme environmental conditions, including simulated seawater, high humidity, high/low pH, very low temperature, and moderately high temperature, this adhesive exhibited strong lap-shear strength. In comparison to several widely used commercial adhesives, our adhesive showed either superior or competitive lap-shear strength and toughness. This superior performance arose from the synergistic interplay of dynamic covalent and noncovalent interactions both within the adhesive network and at the substrate interface. DFT calculations revealed that the facially amphiphilic molecular design enabled electronegative atoms to preferentially orient toward substrate surfaces, promoting multiple interactions (e.g., hydrogen bonding and van der Waals) across interfaces. The incorporation of dynamic covalent bonds conferred key advantages: reversible adhesion without performance degradation over multiple cycles, facile removal from surfaces, and chemical recyclability—features absent from most commercial adhesives. Furthermore, by tuning the cross-linker concentration, we achieved pressure-sensitive adhesion with good lap-shear strength on both hydroxylated and fluorinated surfaces.

A preliminary LCA suggests that these adhesives are environmentally competitive with current alternatives. Further reduction in the carbon footprint could be achieved by substituting the JAT403 component with bioderived amines. In summary, this work demonstrates the successful upcycling of postconsumer PET plastic waste into a versatile, high-performance, reversible, and sustainable adhesive. The adhesive's superior performance in both dry and wet environments, combined with its reversible and recyclable nature, presents a sustainable solution for plastic waste management and offers commercial potential.

MATERIALS AND METHODS

Synthesis of the amphiphilic PET macromonomer

An amphiphilic PET macromonomer was synthesized through the deconstruction of PET via catalyst-free aminolysis. PET pellets (molecular weight of 60 kg/mol; 10 g, 0.096 mol of esters) were added to a 100-ml vessel containing a stir bar and 5 equivalents by mass (3.5 mol equivalents relative to esters) of JAT403 (50 g, 0.34 mol of amines). The mixture was stirred at 180°C for 7 hours, or until the PET pellets were completely dissolved, resulting in a light-yellow viscous solution. The reaction mixture was distilled to remove excess ethylene glycol before being transferred in portions of ~ 5 ml into eight 50-ml centrifuge tubes. The remaining 45 ml was filled with hexanes, and the tubes were vortexed and shaken to suspend the precipitated product. The tubes were centrifuged, and the supernatant was decanted off. This process was repeated five times. The PET-JAT403 macromonomer product was collected as a viscous liquid and dried overnight in a vacuum oven at 100°C to yield product (100%).

Synthesis of the amphiphilic TCDAcAc cross-linker

TCD diol (20 g, 0.10 mol) was added to a 100-ml round-bottom flask with excess *tert*-butyl AcAc as solvent (160 g, 1.0 mol). The reaction mixture was refluxed at 150°C for 5 hours before increasing the temperature to 150°C and distilling unreacted *tert*-butyl AcAc. After 5 hours, the product was suspended in hexanes and allowed to settle before decanting off the excess solvent. The product was dried at 80°C in a vacuum oven for 16 hours to yield the TCDAcAc product (66% yield).

Preparation of the reversible TCD-V adhesive

The PET macromonomer (0.5 g, 1.8 mmol of amine) was weighed into a 1-dram (3.7 ml) scintillation vial before adding TCDAcAc in the desired ratio (0.25 g, 1.4 mmol for 10:7) directly to the vial containing the PET macromonomer—no solvent was added. The amount of TCDAcAc was varied to accommodate the desired amine:AcAc ratio. After adding the cross-linker, the mixture was vigorously stirred for 30 s, scraping the vial walls to ensure complete mixing and yield the reversible TCD-V adhesive.

Lap-shear adhesion testing

Lap-shear adhesion measurements for Al, steel, and wood were conducted following a modified version of the ASTM D1002 method (ASTM D1002-10) in an Instron tensile frame equipped with a 5-kN load cell at a crosshead speed rate of 10 mm/min. Lap-shear adhesion strength (in megapascals) was calculated from the force versus extension curve as the maximum force (in newtons) divided by the overlap area (in square millimeters). The work of debonding (in

newtons per meter) was calculated by taking the area under the force versus extension curve and dividing by the surface area of the adhesive.

Underwater application

For underwater adhesion testing, ~100 μl of adhesive mixture, loaded into a 1-ml syringe, was applied to a submerged substrate with an area of ~144 mm^2 (12 mm by 12 mm) in RT water. A second substrate was then pressed onto the adhesive, and the assembly was allowed to cure underwater overnight (16 hours) at RT before lap-shear strength testing.

Structural application

For the structural liquid adhesive, the adhesive mixture was spread onto the substrates at RT and cured at elevated temperature (80°C) for 6 hours to maximize adhesion. After cooling, the substrates were tested for lap-shear strength. For the dry (melt) adhesive, pieces of fully cured TCD-V film (6 mm by 6 mm) were placed between overlapping substrates and hot-pressed at 120°C for 30 min under constant pressure [500 psi (3447 kPa)]. After cooling to RT, lap-shear strength was measured ($n = 3$; reported as means $\pm$ SD).

Pressure-sensitive application

For pressure-sensitive applications, the 10:0.5 TCD-V adhesive was mixed and cured at 80°C for 1 hour before applying to the substrates and pressing with 800 g of pressure through application of a metal weight for 10 min. Alternatively, the lap-shear strength could be increased by applying large amounts of pressure [145 psi (1000 kPa)], as shown in fig. S19.

DFT calculations

DFT calculations were used to investigate the mechanism underlying the versatile, tough, and reversible adhesion of the TCD-V adhesive. A hydroxylated silica surface was used in the calculations to model interactions, given the significance of dynamic bonding as the primary energy contribution. Further details regarding the DFT calculations can be found in the Supplementary Materials.

Supplementary Materials

This PDF file includes:

Supplementary Text
Figs. S1 to S29
Tables S1 to S11

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High-performance reversible adhesive from PET waste for underwater, structural, and pressure-sensitive applications

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